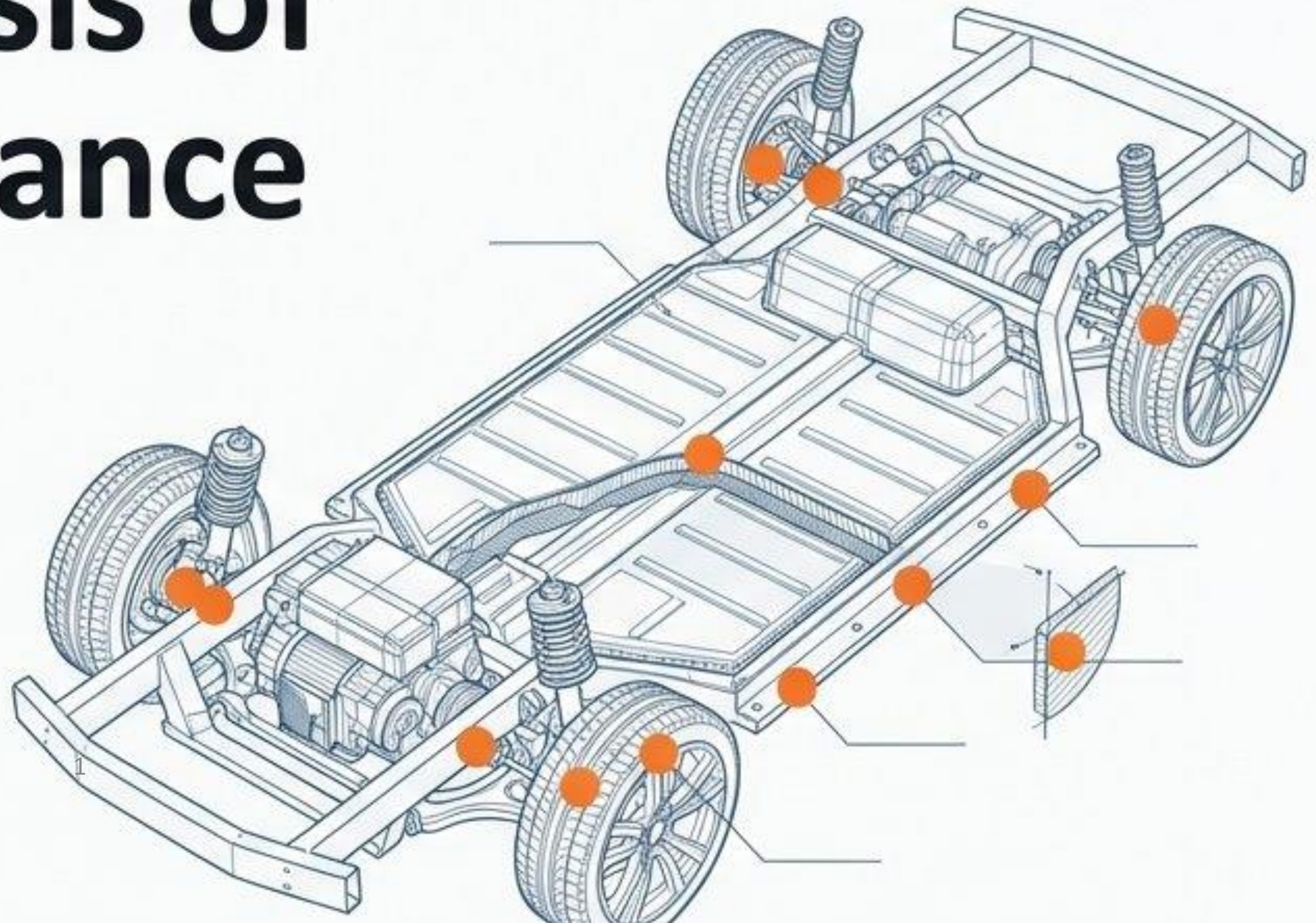
Quantitative Analysis of Predictive Maintenance for Electric Vehicle Components



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MBAR 661 Capstone Research Project
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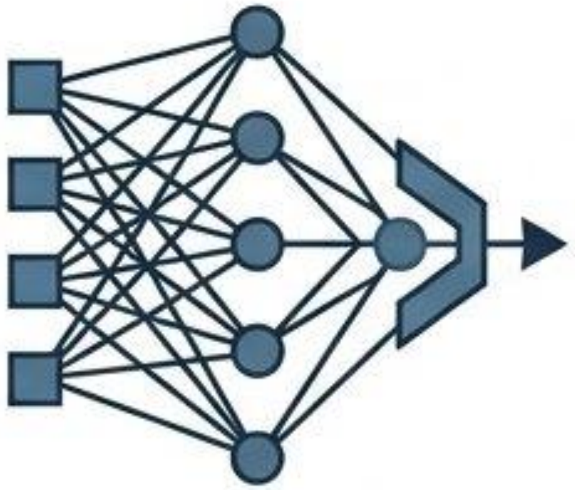
Land Acknowledgement

We acknowledge that the territories on which UCW and its campuses are situated are the traditional, ancestral and unceded territories of the xʷməθkʷəy̓əm (Musqueam), Sḳwx̱vú7mesh (Squamish) and Səlílwətał (Tsleil-Waututh) Nations. We are grateful for the opportunity to live, work, and play on these lands and to the Indigenous Nations who have cared for the land since time immemorial.

What is Predictive Maintenance

Operational data collection

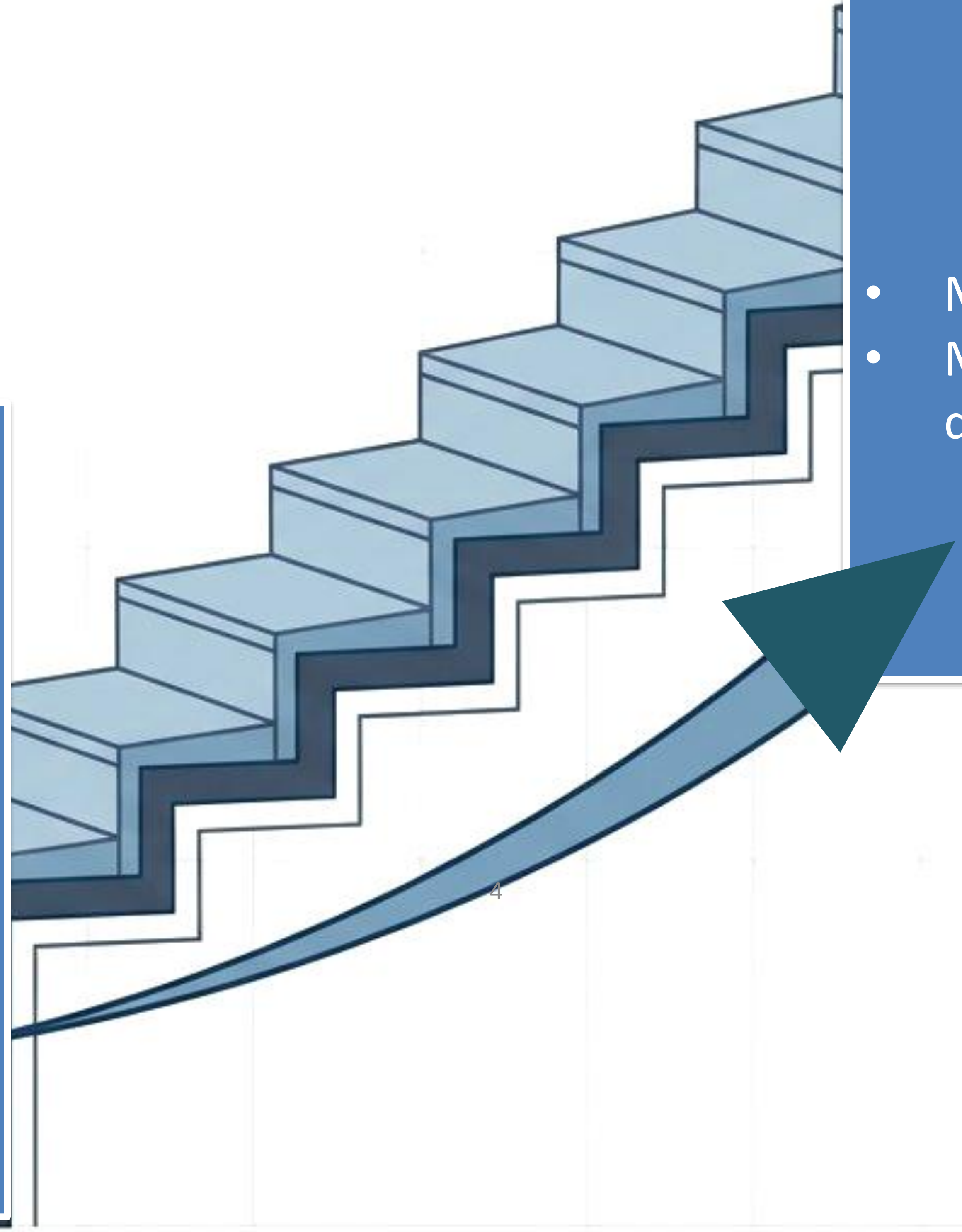
Analytics detect Abnormal patterns



Machine Learning models

Proactive Maintenance

Why Predictive Maintenance Matters



Cost Centre

- EV TCO vulnerable to unplanned Maintenance
- Battery and Electric components Risks

Value Driver

- Maintenance a Strategic capability
- Mitigate Emergencies and Vehicle downtime

Why Traditional Maintenance Fail

- Over reliance on static mileage or temporal internals
- Blindness to rapid degradation within high stress environment
- Resource waste on unnecessary premature preventive checks
- Severe misalignment between maintenance schedules and physical component wear.

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Traditional Maintenance vs Predictive Maintenance

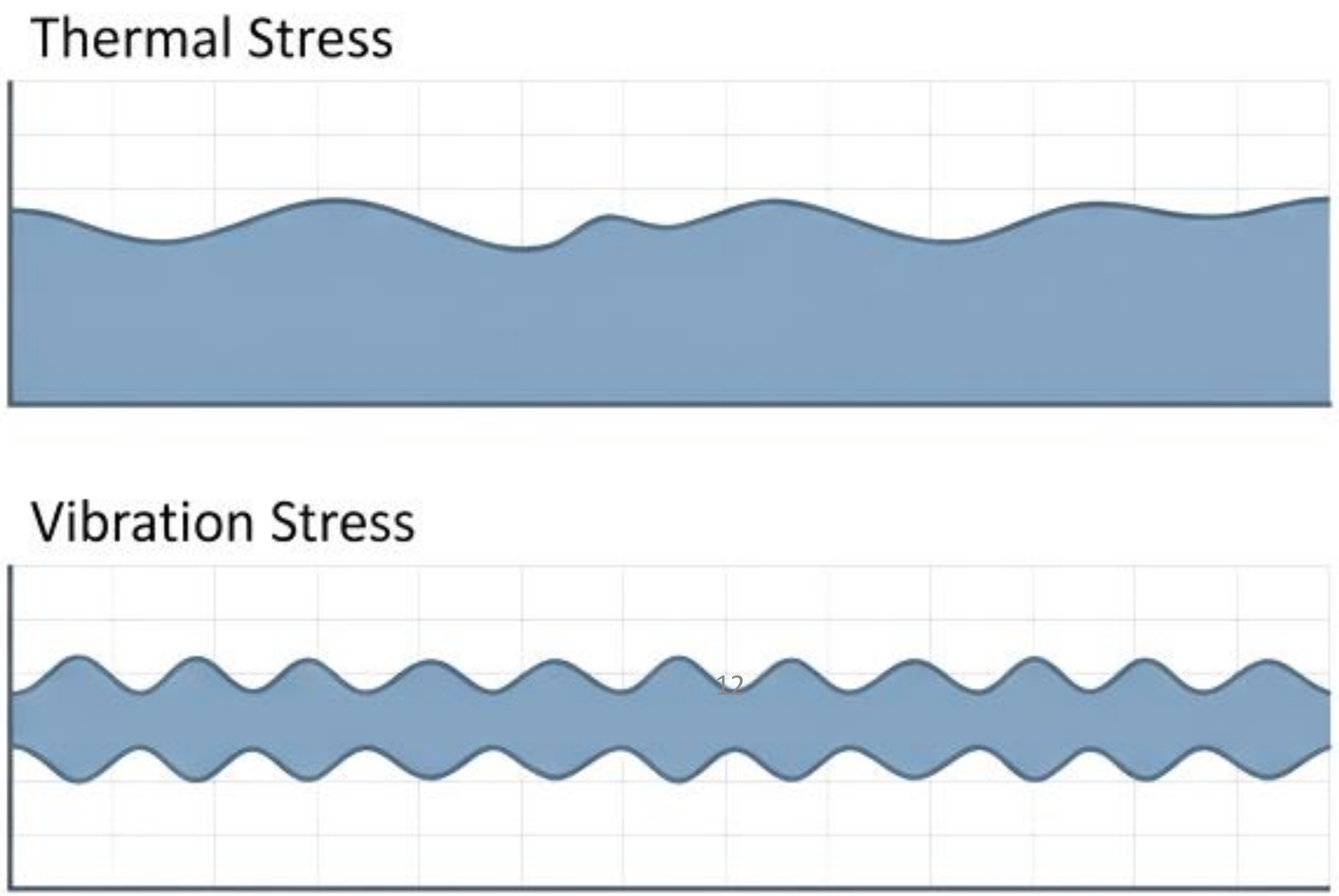
	Traditional Preventive	Predictive
Trigger Metric	Mileage / Time intervals	Component Stress / RUL
Environmental Factors	Additive / Isolated	Synergistic / Combined
OPEX Profile	High Unplanned Spend	Optimised Planned Spend
Failure Risk	Vulnerable to 'Failure Tail'	Mitigated Tail Risk

EV degradation highly conditional. Mileage or Time not enough.

Research Problem

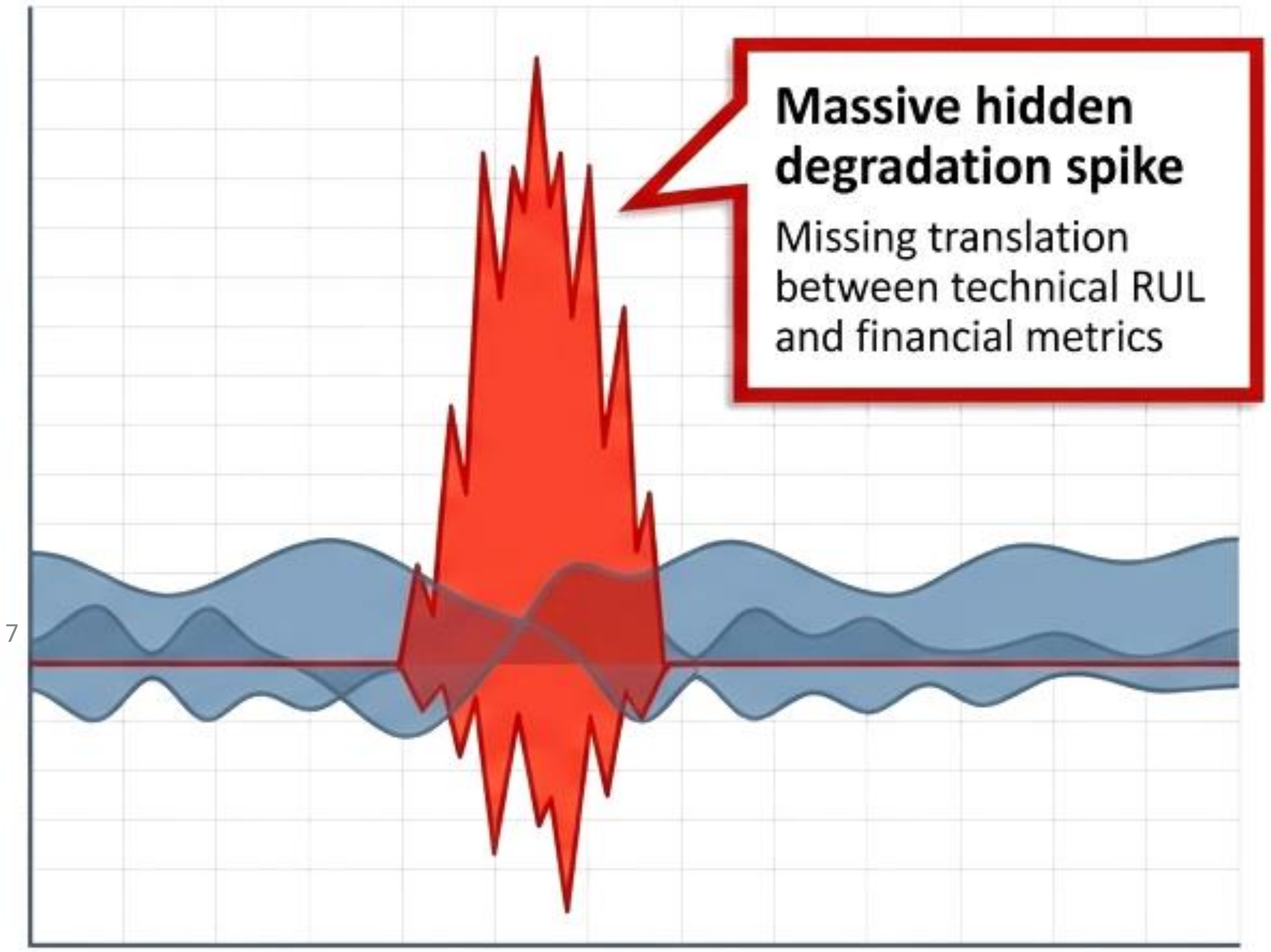
Industry standard: Isolating route roughness and battery temperature fails to account for synergistic environmental effects.

Isolated Variables



Underestimation of failure risks

Combined Synergistic Stress



Research Goal(s)

- To verify the synergistic environmental stressors
- Achieve a robust RUL prediction estimation
- Develop a context-aware predictive model
- Establish a direct quantification of TCO Impact

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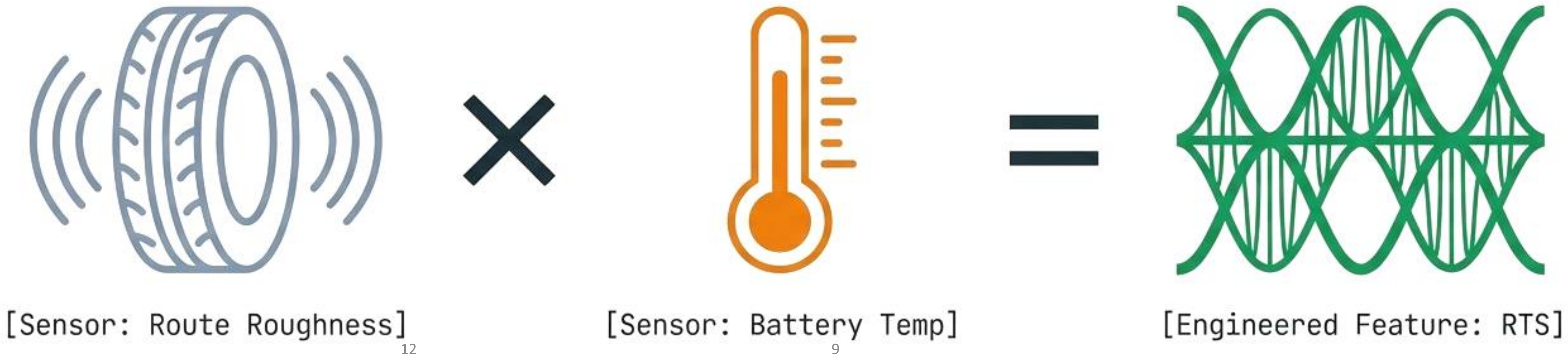
Engineer the Roughness-Temperature Synergy RTS



RTS = Route Roughness X Battery Temperature

Machine Model Performance

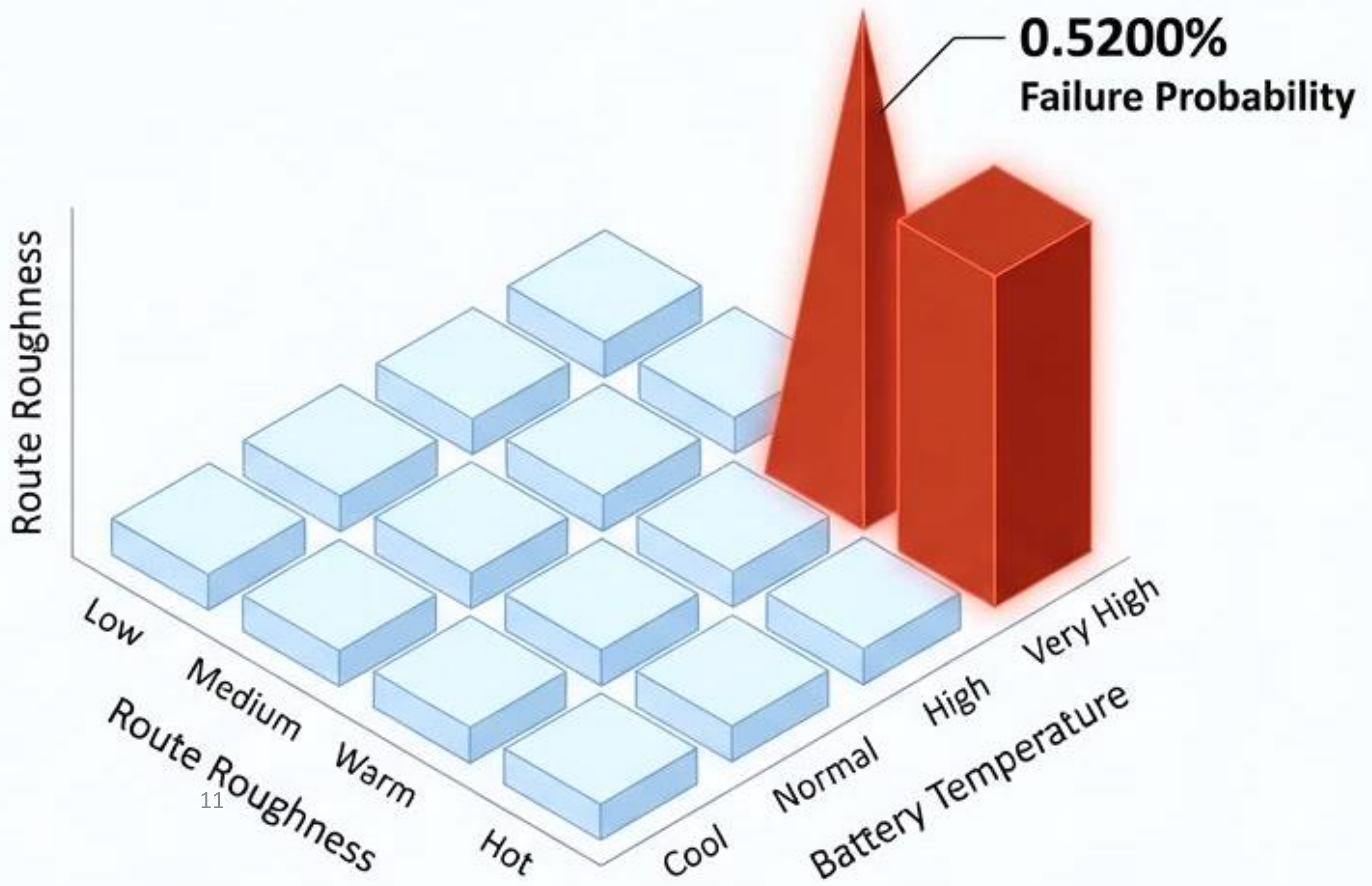
Model Architecture	Non-linear Capture	Synergy Sensitivity	Recall Potential
Linear Regression (ElasticNet)	Low	Low	0.08%
Polynomial (Degree 2)	Moderate	Moderate	Overly sensitive to sensor noise
Gradient Boosting Machine (GBM)	**High**	**Highest**	**84.8% Target Recall**

Key Findings

The 226x Synergistic Multiplier

Baseline:
0.0023% Failure Probability.

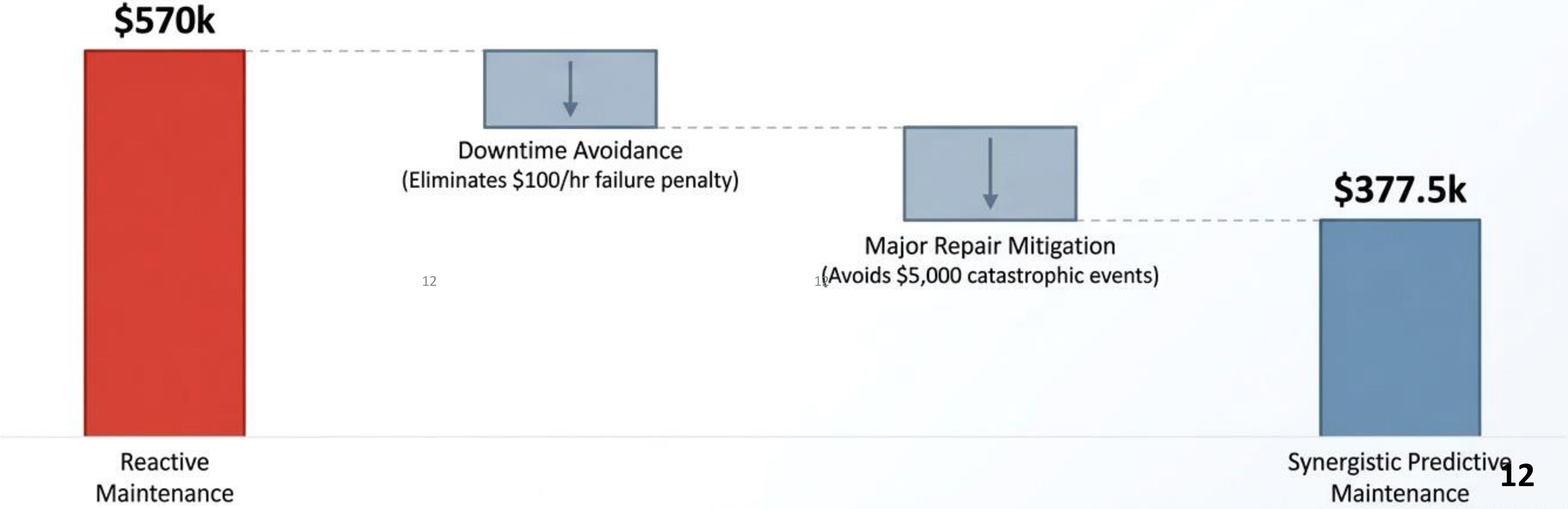
Synergistic Stress Zone:
0.5200% Failure Probability.



Financial Impact

33.8% Total Annual OPEX Reduction

160.4% First-Year Return on Investment (ROI)



Strategic Fleet Implementation

- Transition to fully stress-based maintenance scheduling
- Deploy Synergy-Aware Routing systems
- Divert assets from rough terrain during high thermal loads
- Optimise inventory via predictive just-in-time ordering

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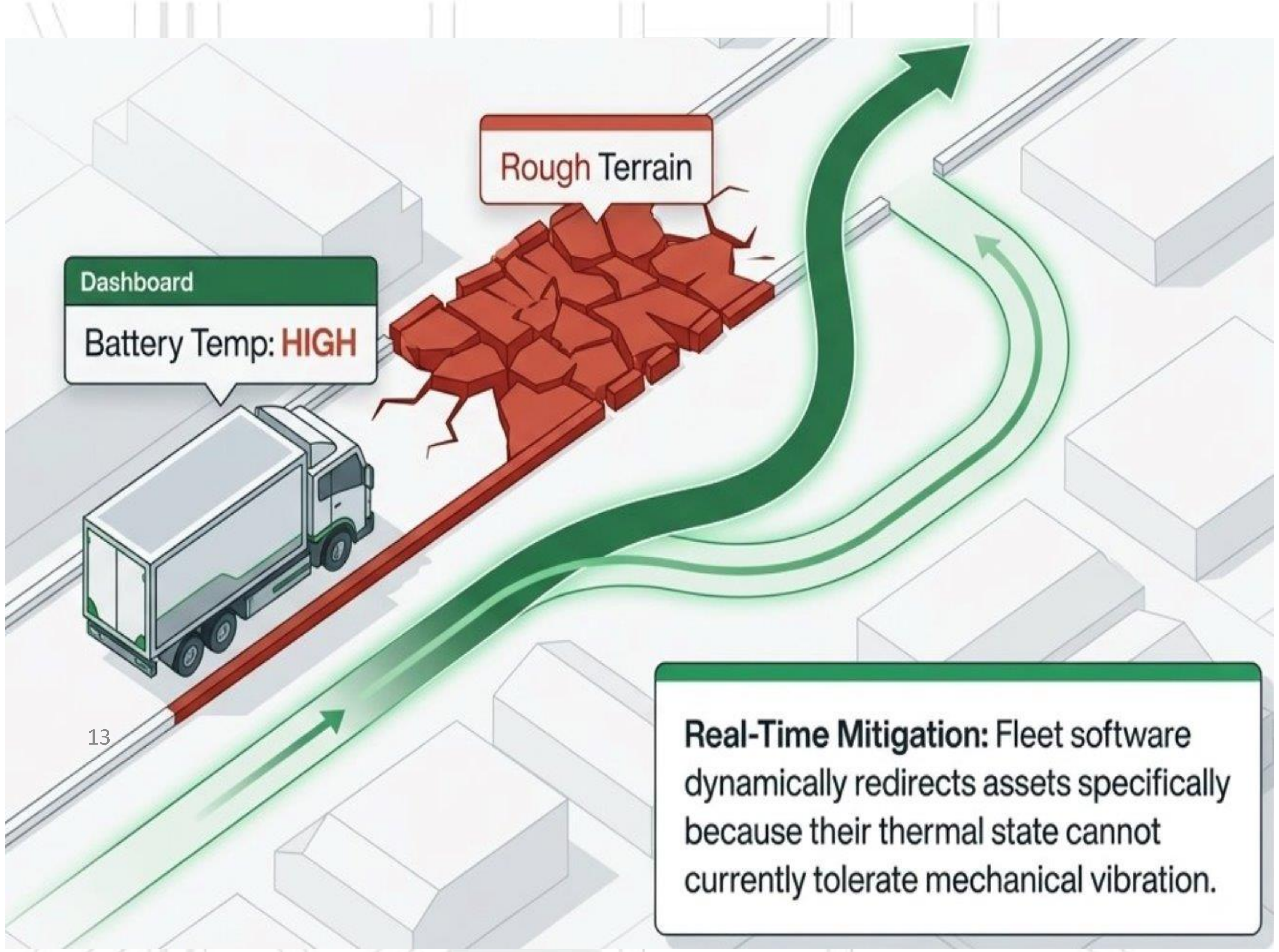

Conclusion

- Roughness-Temperature-Synergy's impact 
- GBM Multiplier effect 
- Economic Value on OPEX and ROI 
- Impact on the Market 

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Strategic Alignment with Canadian EV Market

2026 $2.3b EV Market
2035 EV Mandate

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New 6.1% International Tarriff structures

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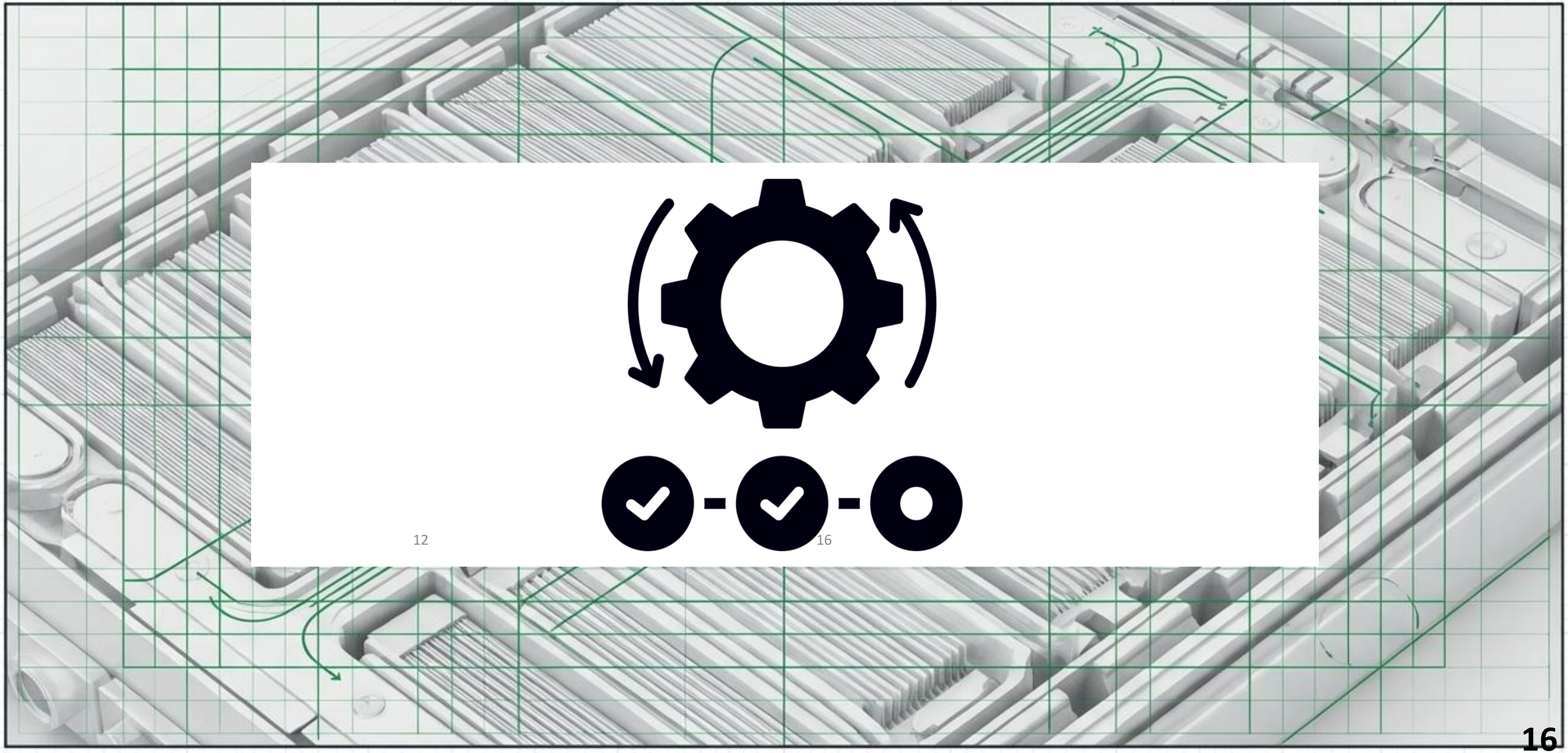
Revolution in acturial pricing

Operational Intelligence is the Ultimate Asset



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Thank You For Your Attention

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Question?